Computational Platform for Synthetic Correction of Disease-Causing SNPs: Sickle Cell Disease

Luke Hovagimian, Alexander Leong, Minh Nguyen, Heewon Park
Point of Contact: lukehov@bu.edu

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Goal & Motivations

SNPs contribute to various diseases and health conditions, but **current methods focus mainly on detecting these mutations** rather than developing therapeutic solutions. A **computational framework** is needed to **identify SNPs** and guide personalized **genetic correction strategies**.

GOAL:

1) Identifies disease-associated SNPs from patient sequences (**HBB**)
2) Generates Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) based correction strategies
3) Evaluates guide ribonucleic acid (gRNA) based on efficiency and off-target safety

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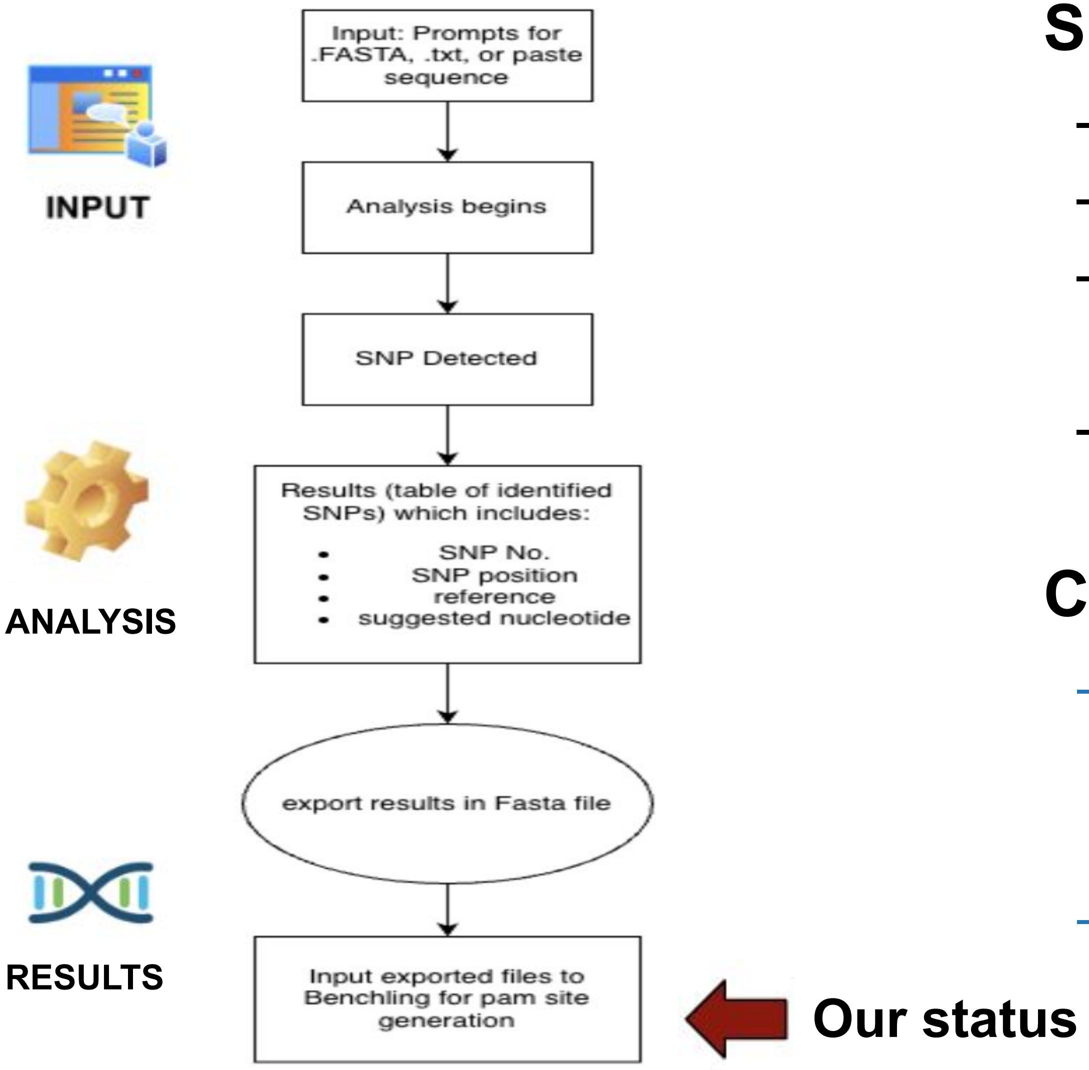
General Functionality Workflow



SNP detection:

- user input
- determine if SNP is detected or not
- shows results including the SNP number, position, reference and suggest nucleotide.
- export the results in fasta file

CRISPR gRNA:

- <u>Fasta file export</u>
  - input exported Fasta file to benchling
  - pam sites generated by benchling
- <u>gRNA and CRISPR embedded function</u>
  - output results similar to the FASTA file are inputted and gRNA design is created

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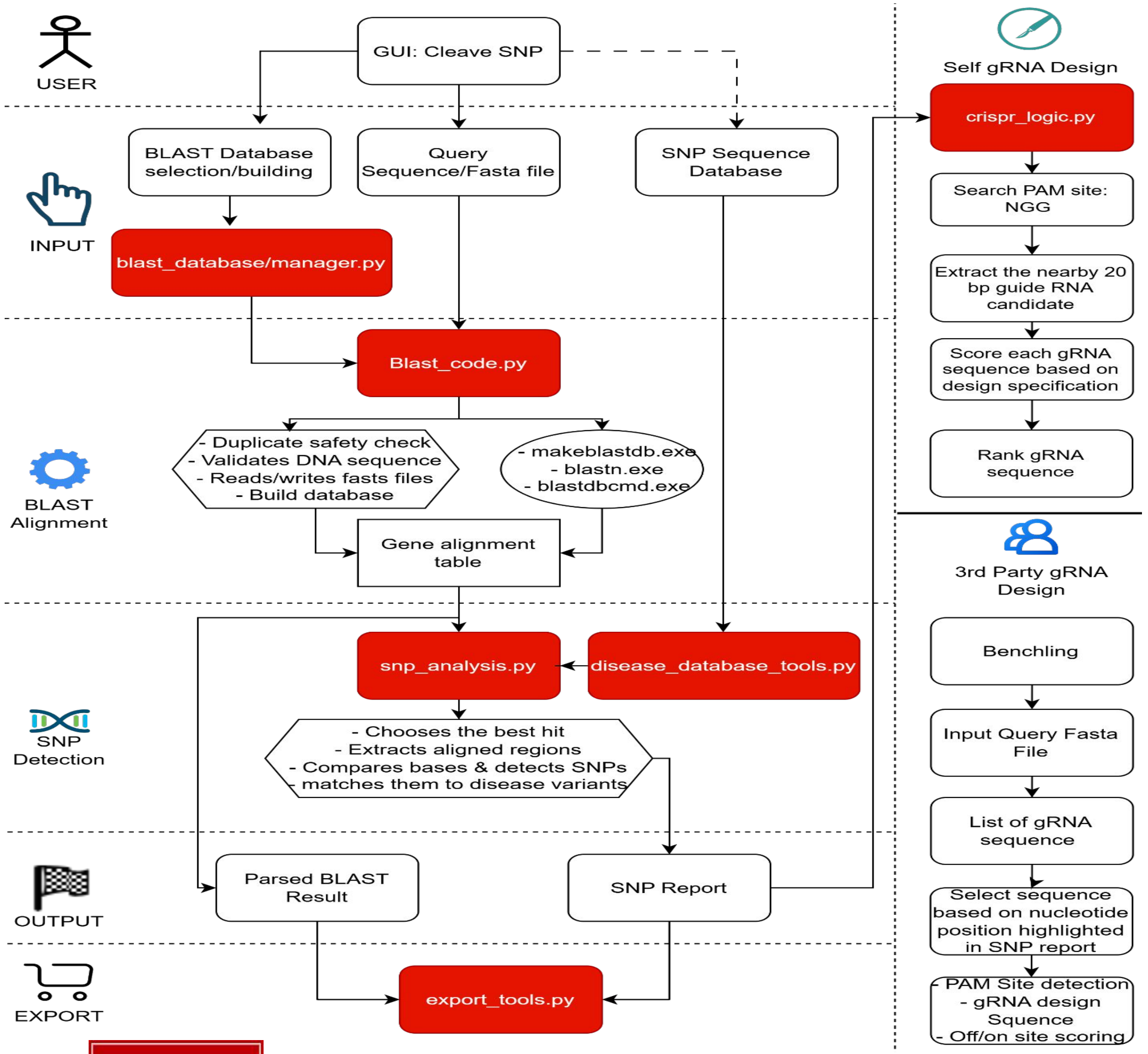
Design Specification

Design	Specification
Input query sequence handling & capacity	Accept FASTA files or manual gene-sequence entry; only valid nucleotide characters (A, C, T, G, N) are allowed; **minimum query length: 20 bp**
Database handling & safety check	Accept FASTA files or manual gene-sequence entry; only valid nucleotide characters (A, C, T, G, N) are allowed; **reject malformed or duplicate records of >99% similarity**; database sequence should be longer than the query for reliable alignment
SNP analysis accuracy	Select high-quality alignments using mismatch presence, percent identity, and alignment logic; **detect the mutated nucleotide position with 100% accuracy on validated test cases**
gRNA Design & Scoring	gRNA design with a valid PAM site, scoring (0-1.0) based on **minimal SNP-to-cut-site distance (30%), 40-65% GC content (25%), low homopolymer penalty (15%), and low off-target activity (30%)**
GUI intuitiveness	Present inputs, actions, and outputs clearly; **complete the workflow in 7 steps or fewer**

Integrated Detailed Workflow



Within our code we are able to generate **GRNA** within our website. However, we need to build more on the **database** to measure its accuracy.

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FASTA file

Exported results in FASTA file:

- Gene Target (e.g., HBB)
- Genomic Coordinates (reference location)
- Mutation ID (e.g., c.20A>T)
- **Target Nucleotide**
- disease label
- **Reference Nucleotide: A (within reference sequence)**
- Homology Arms: Flanking DNA sequences (in both Query and Reference strings)
- Exon/Intron Map: Sequence differences between short query and long reference blocks

```
>HBB_sickle_patient_long|gene=HBB|variant=c.20A>T|disease=Sickle_cell_disease|role=query|snp_count=1
AGTTGGTGGTGAGGCCCTGGGCAGGTTGGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCATGTG
GAGACAGAGAAGACTCTTGGGTTTCTGATAGGCACTGACTCTCTCTGCCTATTGGTCTATTTTCCCACCCTTAGGCTGCT
GGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAACCCTA
AGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCTCAAGGGCACCTTT
GCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGATGGTGCATCTGACTCCTGTGGAG
AAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGA
>NC_000011.10:c5227071-5225464 HBB [organism=Homo sapiens] [GeneID=3043] [chromosome=11]|role=reference_best_hit|snp_count=1
```

Benchling CRISPR Functionality & Specificity Scoring

Purpose: Import a FASTA file output from web app for running a CRISPR Design Analysis to create a list of viable PAM sites with On/Off-Target Scores and export list as a table to visualize and chose PAM site

Specificity/Efficiency Scoring:

- **Specificity Score:** how likely the guide is to cut somewhere else in the genome by mistake.
- **Efficiency Score**: How efficiently your guide will actually cut at the intended site
- **Goal**: From the list of PAM sites, find a site that sits a decent percentages for both (aim for both to be around or above 60%)

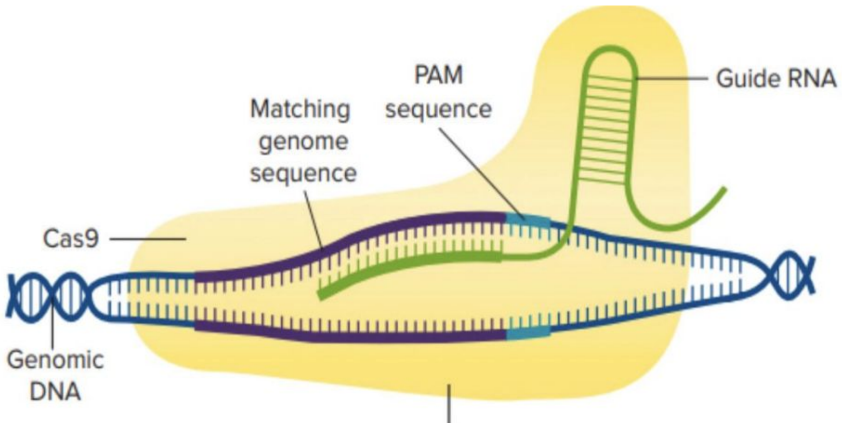
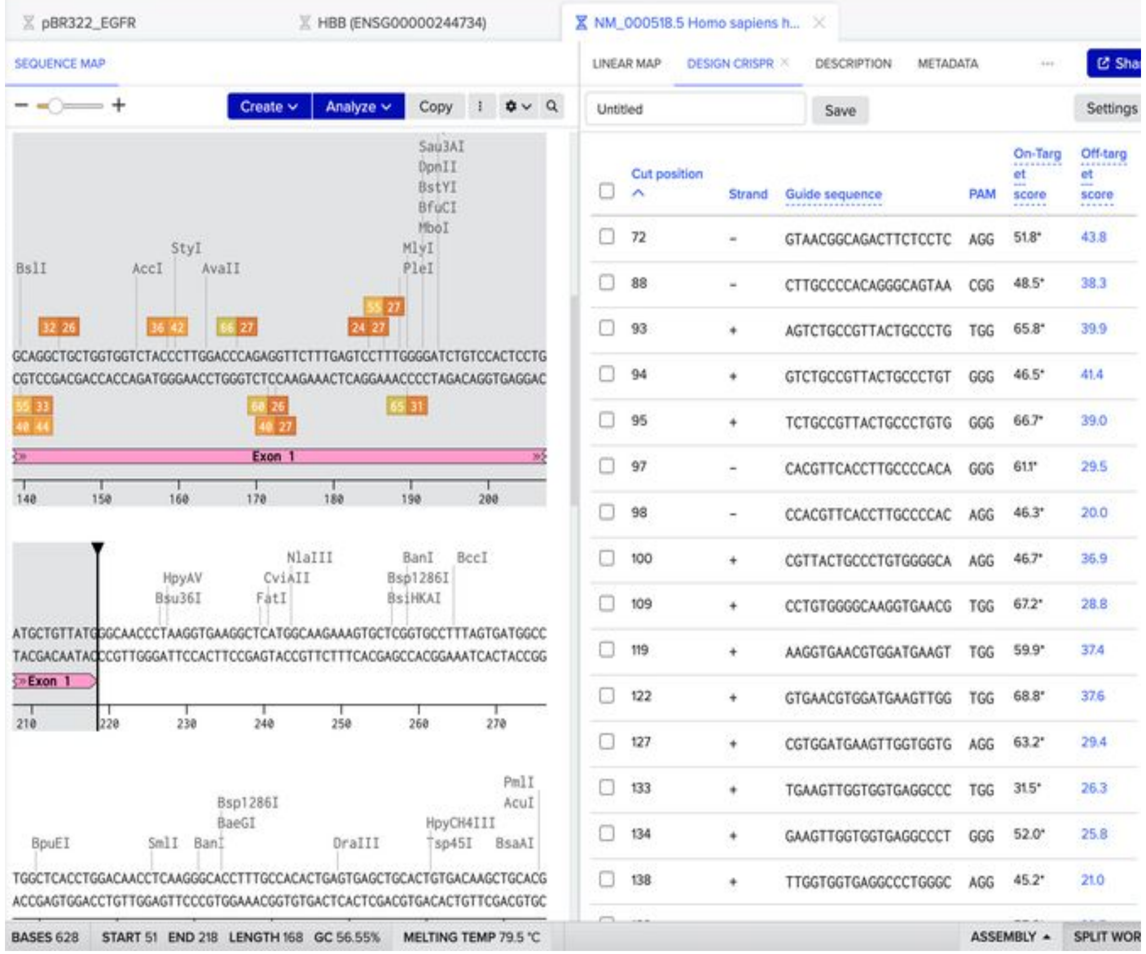


Image: https://platform.cysf.org/media/CRISPR_Cas9_-_The_genetic_scissors_to_treat_Sickle_Cell_Disease1.pdf



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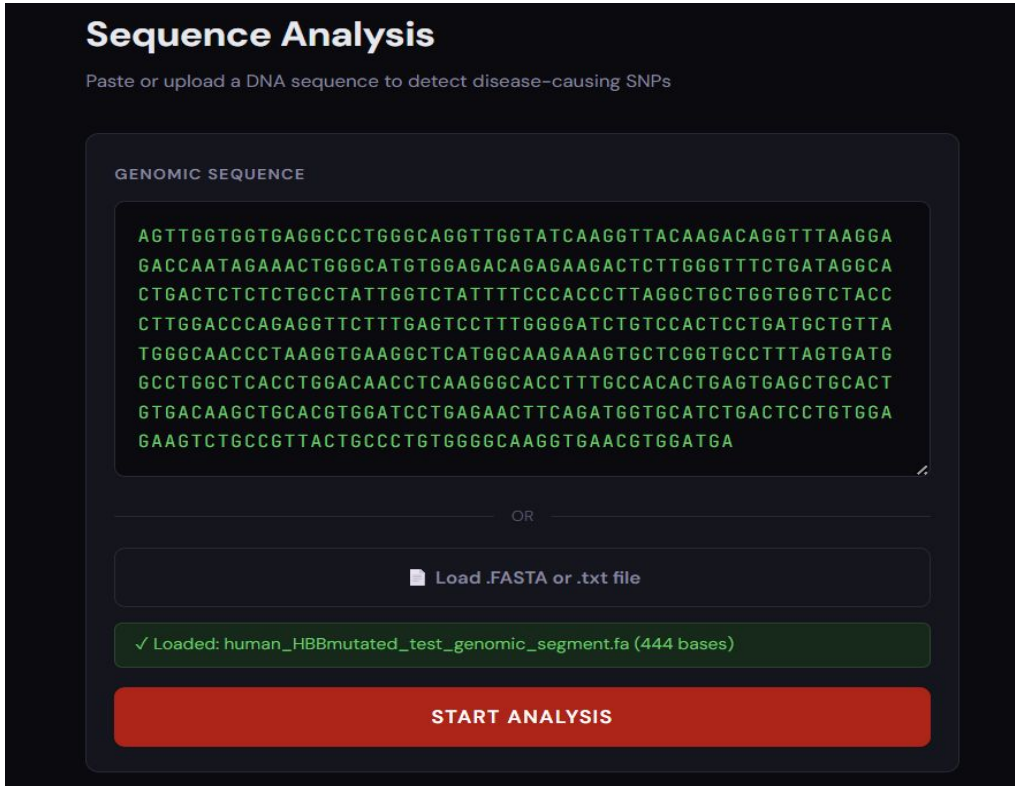
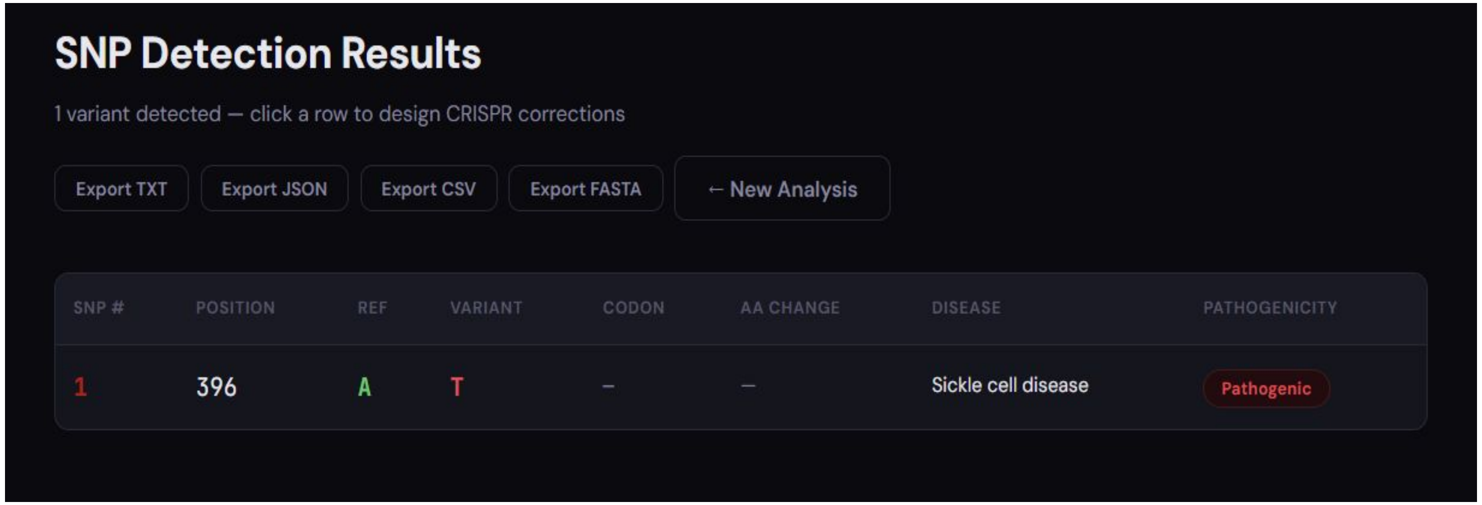
Technical Example – SNP detection





Identified position off by 1 due to python indexing that starts from 0 instead of 1

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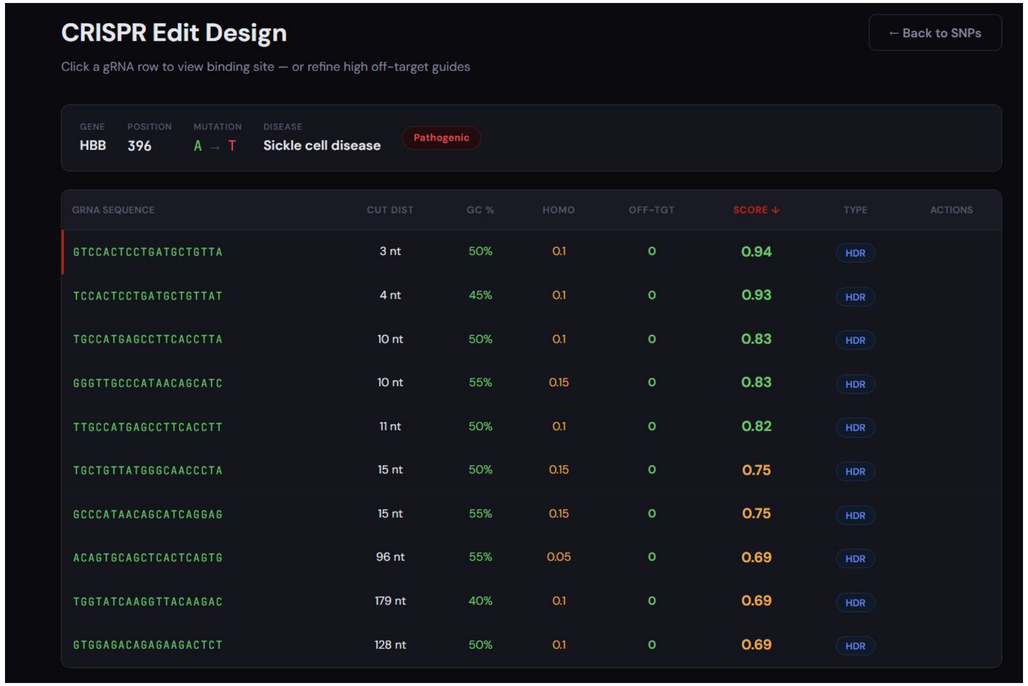
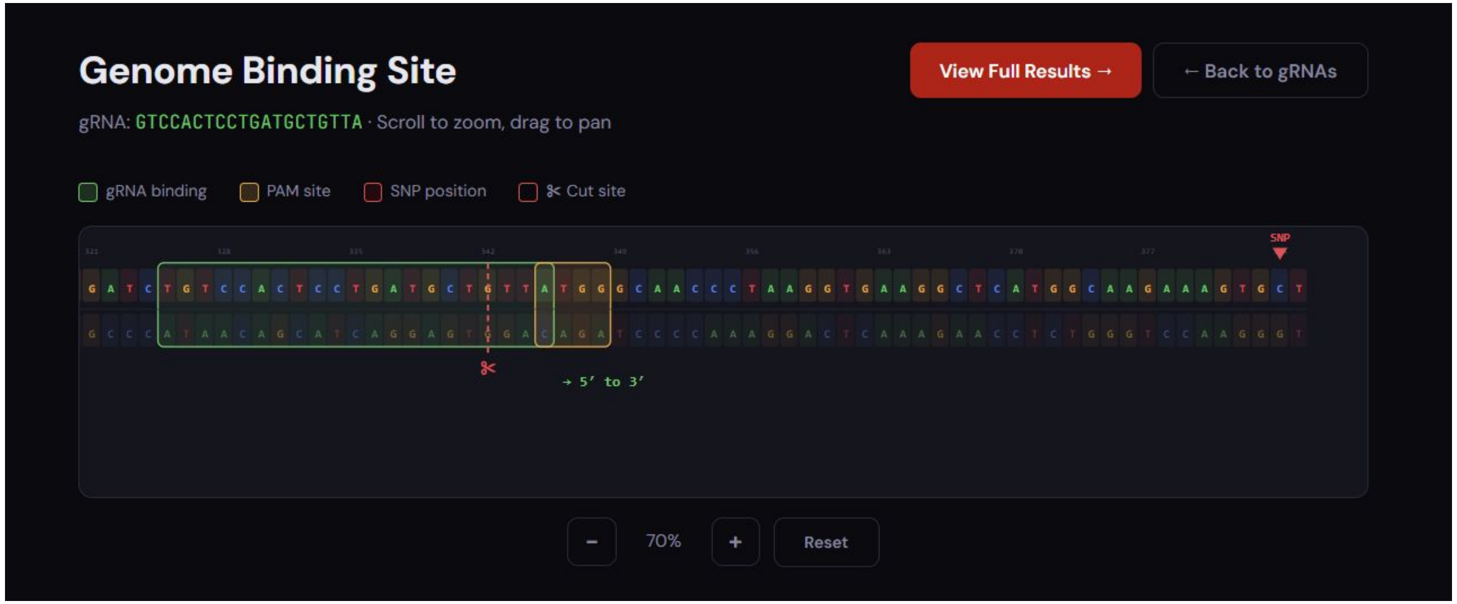
Technical Example – gRNA design





More calibration and tuning needed

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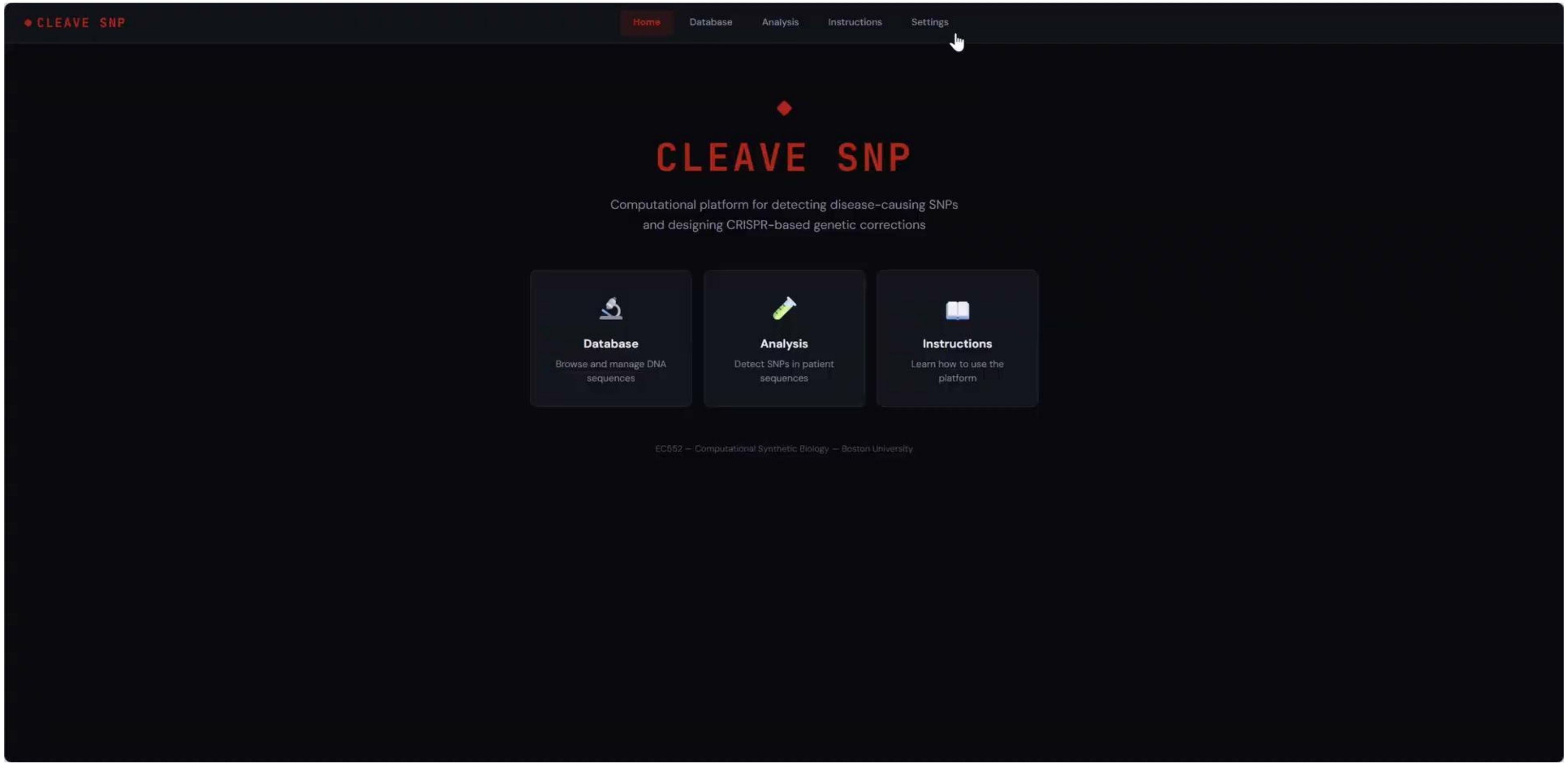
Demo movie



It takes 6 steps:

1) Configure Blast Bin
2) Build Database
3) Input query fasta
4) Start SNP analysis
5) gRNA design generation
6) Visualization of gRNA binding

Our GUI meets the current **intuitiveness expectation of <7 steps**; having minimal and more concise steps.

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https://drive.google.com/file/d/1Zlv9HJRwwX3o1XI_NMbALaPO3bRwJb7A/view?usp=drive_link

Failures

Failure	Impact	Potential Solution
BLAST alignment ambiguity/uncertainty	Gaps and shifted boundaries caused SNP-analysis errors	Improve alignment filtering and comparison logic
Reference & SNP disease database dependency	Missing metadata led to unknown variant matches	Standardize and validate sequence records; and
Improvised gRNA workflow	Pivoted to Benchling: guide scoring and visualization were built but not fully tested	Add automated Benchling export parsing and ranking
Self-coded gRNA design module was not validated	Guide sequence, ranking and specificity could not be fully trusted yet	More tuning and calibration needed for gRNA generation & validation. To be assessed against gold-standard gRNA design tools

Connection to Synthetic Biology Field

1) Mutation analysis and SNP identification
2) Target definition for genome editing
3) Support for downstream gRNA design
4) Future expansion into circuit design, assembly planning, and protocol design

Foundational core of this project is to serve as mutation-analysis and target-definition tool that bridges sequence analysis with downstream synthetic biology design

THANK YOU!

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